

EmoMV: Cross-Modal Emotion-Conditioned Music Generation

Pipeline, Architectures, and Experiments

EmoMV Project

July 7, 2026

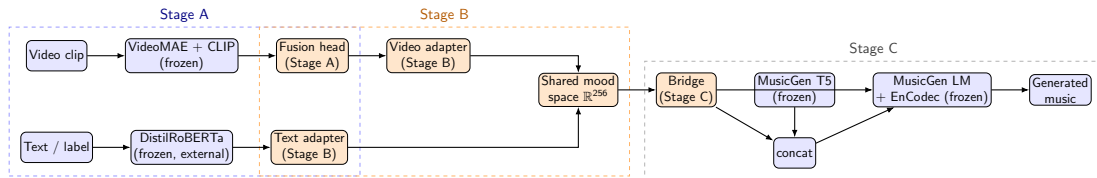
Motivation

- **Goal:** given a video clip (or a text prompt) expressing a mood, generate music that matches that mood.
- **Constraint:** no fine-tuning budget for any large pretrained model — not the video/text encoders, not the music generator.
- **Constraint:** no instance-level (video, caption) pairs exist — only shared categorical mood labels across independently collected video and text data.

Approach

Keep every large pretrained model **completely frozen**. Train only small adapters that translate between representations: video/text $\rightarrow$ shared mood space $\rightarrow$ music-generator conditioning.

Pipeline Overview



Blue = frozen pretrained model

Orange = trained in this project

Dataset & Preprocessing

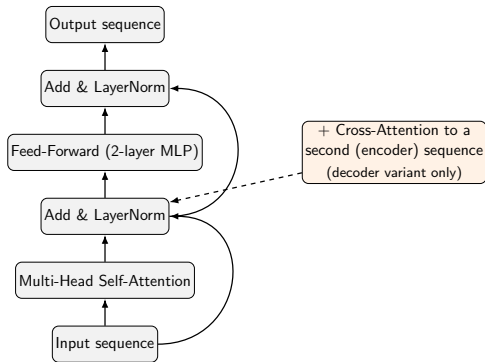
EmoMV dataset (DS1/DS2/DS3), 5 moods: exciting, fear, tense, sad, relax.

- MATCH-only clips kept (video mood = audio mood by construction) — MISMATCH dropped.
- 34 exact duplicates removed (SHA-256), 2 label conflicts resolved.
- 20 genuine duration outliers ($>35s$) windowed into 30s chunks.
- **Final manifest: 3,189 rows**, 3,144 unique videos.

Official split leaked (same source video's segments across train/val/test) — rebuilt via grouped, stratified greedy assignment:

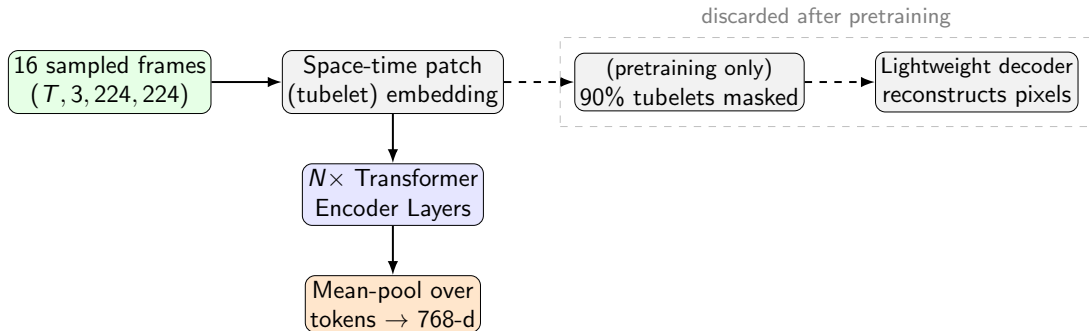
Split	Rows	DS1	DS2	DS3
train	2,232	1,870	190	172
val	479	395	59	25
test	478	388	59	31

Building Block: The Transformer Layer



Stacking N of these (encoder: self-attention only; decoder: self- *and* cross-attention) is the building block behind every transformer used or built in this project — VideoMAE, CLIP’s text tower, RoBERTa, T5, MusicGen’s decoder, and our own bridge. Ref: Vaswani et al., “Attention Is All You Need,” NeurIPS 2017.

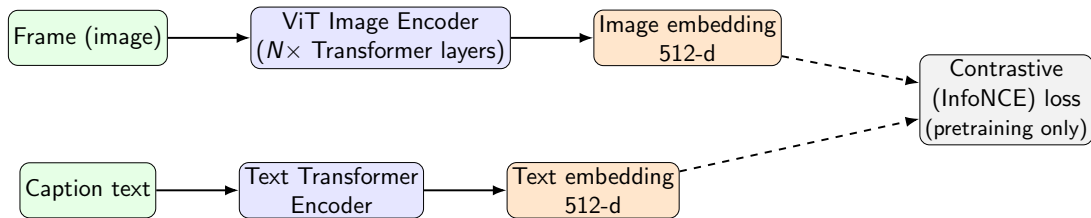
Stage A — Video Encoder 1: VideoMAE



Our usage: MCG-NJU/videomae-base, fully frozen, only the encoder is used — masking/reconstruction was the *pretraining* objective, not something this project runs.

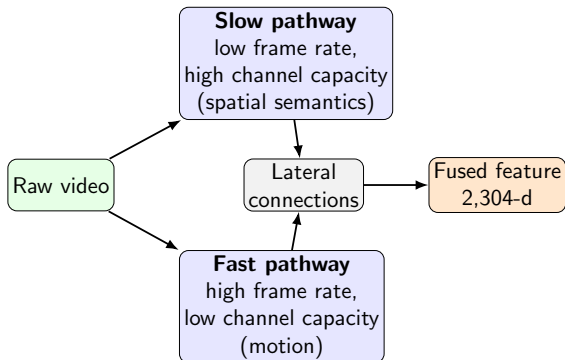
Output: 768-d embedding per clip. Ref: Tong, Song, Wang, Wang, “VideoMAE: Masked Autoencoders are Data-Efficient Learners for Self-Supervised Video Pre-Training,” NeurIPS 2022.

Stage A — Video Encoder 2: CLIP



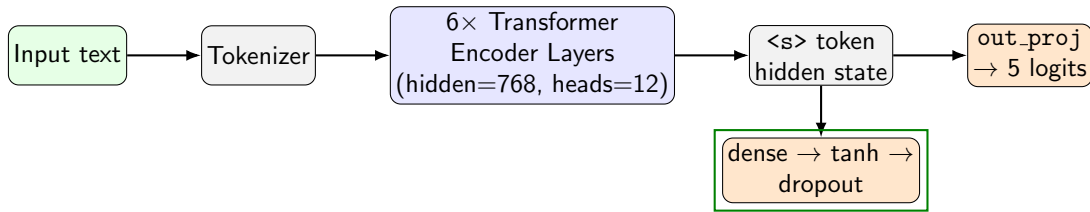
Our usage: openai/clip-vit-base-patch32, fully frozen. Only the *image encoder* is used — per frame, then mean-pooled over 16 frames $\rightarrow$ 512-d. The text tower and contrastive objective are CLIP's own pretraining machinery, not used here. Ref: Radford et al., “Learning Transferable Visual Models From Natural Language Supervision,” ICML 2021.

Stage A — Baseline Video Encoder: SlowFast



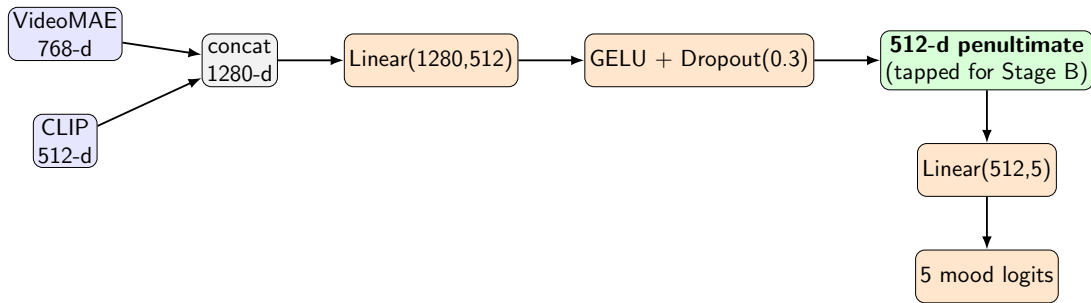
Our usage: not re-run — EmoMV *ships* pre-extracted SlowFast features (2,304-d) per clip; this project only aligns them with its own manifest and trains a small $\text{Linear}(2304, 256) \rightarrow \text{ReLU} \rightarrow \text{Dropout} \rightarrow \text{Linear}(256, 5)$ MLP on top, as a reference baseline the VideoMAE+CLIP fusion model must beat. Ref: Feichtenhofer, Fan, Malik, He, “SlowFast Networks for Video Recognition,” ICCV 2019.

Stage A — Text Encoder: (Distil)RoBERTa



Our usage: an already fine-tuned distilroberta-base 5-class mood classifier (external asset, test macro-F1 ≈ 0.80) is used purely as a frozen encoder. We tap the 768-d *penultimate* representation (green box: dense+tanh, before out_proj), verified to reproduce the model's own logits exactly by manual replay. Ref: Liu et al., "RoBERTa: A Robustly Optimized BERT Pretraining Approach," 2019; Sanh et al., "DistilBERT," 2019 (distillation methodology).

Stage A — Fusion Classifier (this project)



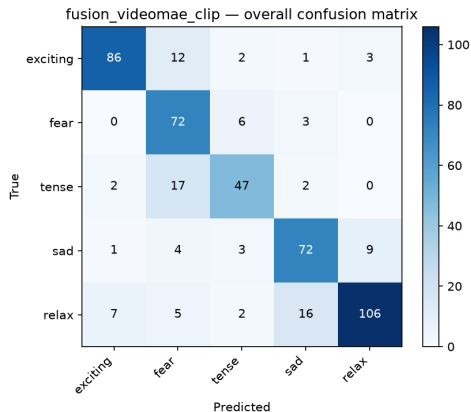
Only this small head is trained; VideoMAE and CLIP remain fully frozen throughout. The green 512-d activation (*after* GELU+Dropout, *before* the final linear layer) is exactly what Stage B's video adapter later projects from — not the raw 1280-d concat, not the logits.

Stage A — Results

Model	Acc	Macro-F1	W-F1
SlowFast	50.6%	0.479	0.503
Fusion (ours)	80.1%	0.794	0.803

	DS1	DS2	DS3
Fusion acc.	86.6%	52.5%	51.6%

A clear domain-shift gap: DS2/DS3 are only 9% of the data and were flagged as distributionally different by the EDA's own KS tests.

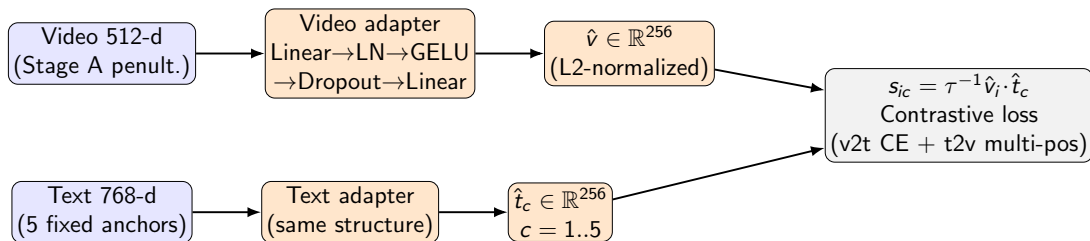


Stage B — The Pairing Problem

- Standard CLIP-style contrastive training needs (image, caption) *instance* pairs.
- We have none: $\sim 3,144$ videos and a text corpus, each independently labeled with one of the same 5 mood classes — **no video $\leftrightarrow$ sentence correspondence**.
- **Solution:** use the 5 class-name strings themselves as fixed *text anchors*, and train with a **label-anchored supervised-contrastive loss** — every video is pulled toward its class's anchor, every anchor is pulled toward all same-class videos in the batch.

Generalizes: Khosla et al., “Supervised Contrastive Learning,” NeurIPS 2020; temperature parameterization follows Radford et al. (CLIP), ICML 2021.

Stage B — Cross-Modal Alignment Architecture

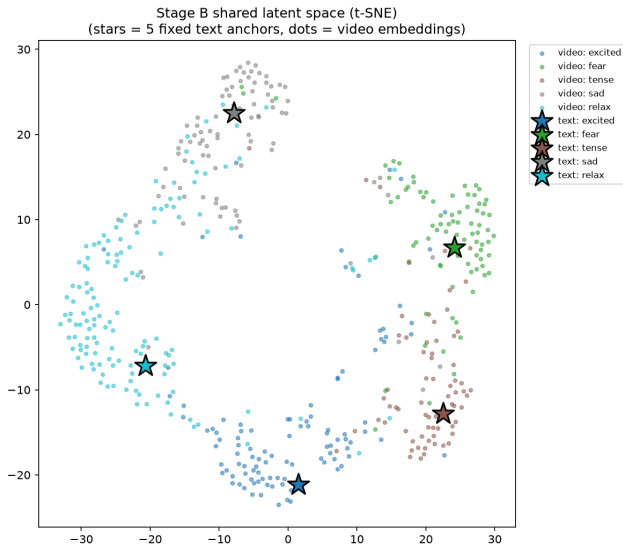


Only the two small adapters (+ a learnable temperature) are trained; the Stage A encoders behind them stay frozen. **Result:** a shared 256-d space where same-mood video/text land close together.

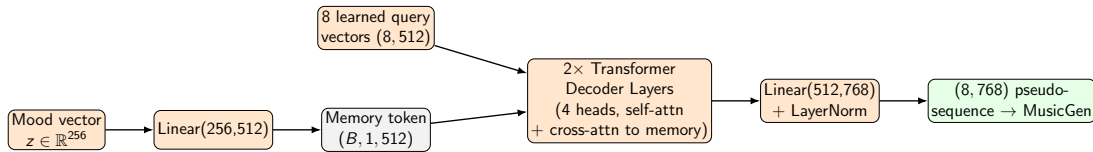
Stage B — Results

Split	V→T Acc.
Overall	80.96%
DS1	88.66%
DS2	50.85%
DS3	41.94%

Modality-gap cosine similarity: 0.69–0.73
across classes — comparable, all positive, no
class catastrophically misaligned.

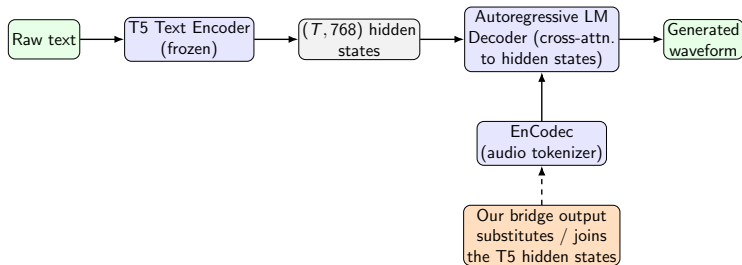


Stage C — The Bridge Network (this project)



Built from the generic Transformer-decoder block: 8 learned queries *cross-attend* to a single memory token derived from z , turning one 256-d point into an 8-token sequence shaped exactly like a T5 conditioning output. Only the bridge is trained; everything it feeds into (MusicGen) stays frozen. Decoder mechanism: Vaswani et al., NeurIPS 2017.

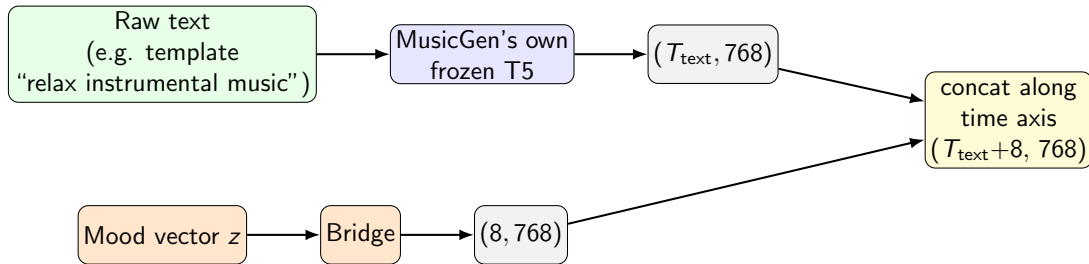
Stage C — The Frozen Generator: MusicGen



facebook/musicgen-small: an autoregressive transformer over discrete EnCodec audio tokens. **Fully frozen** throughout (T5 encoder, EnCodec, LM decoder) — verified against the transformers source: supplying `encoder_outputs` directly bypasses the internal T5 call, exactly the hook our bridge uses. Refs: Copet et al., NeurIPS 2023 (MusicGen); Défossez et al., 2022 (EnCodec); Raffel et al., JMLR 2020 (T5).

Stage C — Milestone 1 Limitation & Dual Conditioning

Milestone 1: bridge output *replaces* T5 conditioning outright — `encoder_outputs=(bridge(z),)`. Sufficient for *which mood*, but discards genre, instrumentation, tempo — anything else in the input text.



Only the bridge is retrained under this new objective (`bridge_dual_best.pt`) — T5 stays frozen (no gradient), and the old single-source checkpoint is kept as a separate ablation baseline.

Stage C — Three-Way Ablation

For every test example (text-anchor *and* held-out video), three conditioning variants are generated and compared by listening:

Variant	Conditioning	Role
t5only	$\text{cond} = T5(\text{text})$	stock MusicGen, zero training
bridgeonly	$\text{cond} = \text{bridge}(z)$	original Milestone-1 design
combined	$\text{cond} = [T5(\text{text}); \text{bridge}(z)]$	proposed dual conditioning

- 40 .wav samples generated across $5 \text{ emotions} \times \{\text{text, video}\}$ sources.
- **No automatic acoustic metric exists** (no FAD/CLAP) — evaluation is deliberately qualitative: does *combined* keep bridge-only's mood consistency while adding audible genre/instrumentation detail from T5?

Cross-Cutting Finding: Domain Shift

	DS1	DS2	DS3
Stage A (fusion) accuracy	86.6%	52.5%	51.6%
Stage B video→text accuracy	88.7%	50.9%	41.9%

- DS2+DS3 are only 9% of the manifest (287/3,189 rows).
- The EDA's own KS tests found DS1/DS2/DS3 clip-duration distributions significantly different — a genuine domain shift, not a labeling artifact.
- The *same* gap reappears independently in two different trained components — strong evidence it reflects the data, not a single model's quirk.

What Was (and Wasn't) Tried

Experiment	Status	Note
SlowFast baseline	Done	Superseded by fusion
VideoMAE+CLIP fusion	Done	Adopted
weighted_sampler imbalance strategy	Implemented, unused	class_weights used instead
Stage B $D = 512$	Not run	Gated on retrieval plateau (never observed)
Auxiliary CE loss (Stage B)	Not implemented	Recommended as collapse safeguard
Bridge Milestone 1 (single-source)	Done	bridge_best.pt
Dual conditioning (concat)	Done	bridge_dual_best.pt
Real per-clip captions	Not available	Template text used instead
Cross-attention balance diagnostic	Not implemented	Only manual listening done
MusicGen LoRA fine-tuning (Milestone 2)	Not implemented	Needs rented GPU
Full MusicGen fine-tuning (Milestone 3)	Not implemented	Needs more paired data

Limitations & Future Work

- **No automatic Stage C metric** — FAD/CLAP/generated-audio mood classifier would replace “listen manually.”
- **Only 5 fixed text anchors** for Stage B’s text side — no diverse caption corpus, so only coarse mood alignment, not fine-grained semantics.
- **Domain shift** (DS1 vs. DS2/DS3) unresolved — more DS2/DS3 data or domain-adaptation would help.
- **Compute-gated**: any MusicGen fine-tuning (Milestone 2/3) needs GPU time beyond the project’s Apple Silicon hardware.
- **Soundtrack audio** used as-is — may include vocals/dialogue, not pure instrumental beds.

References

- Vaswani et al. Attention Is All You Need. *NeurIPS*, 2017.
- Tong, Song, Wang, Wang. VideoMAE: Masked Autoencoders are Data-Efficient Learners for Self-Supervised Video Pre-Training. *NeurIPS*, 2022.
- Radford et al. Learning Transferable Visual Models From Natural Language Supervision (CLIP). *ICML*, 2021.
- Feichtenhofer, Fan, Malik, He. SlowFast Networks for Video Recognition. *ICCV*, 2019.
- Liu et al. RoBERTa: A Robustly Optimized BERT Pretraining Approach. *arXiv:1907.11692*, 2019.
- Sanh, Debut, Chaumond, Wolf. DistilBERT, a Distilled Version of BERT. *arXiv:1910.01108*, 2019.
- Khosla et al. Supervised Contrastive Learning. *NeurIPS*, 2020.
- Raffel et al. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer (T5). *JMLR*, 2020.
- Défossez, Copet, Synnaeve, Adi. High Fidelity Neural Audio Compression (EnCodec). *arXiv:2210.13438*, 2022.
- Copet, Kreuk, Gat, Remez, Kant, Synnaeve, Adi, Défossez. Simple and Controllable Music Generation (MusicGen). *NeurIPS*, 2023.

Thank you

Questions?